

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

Supplementary Material

001 This supplement records the material supporting the
002 main paper without being required to follow its argu-
003 ment: the region-partition procedure and its sensitiv-
004 ity study, complete audit tables for both tracks at ev-
005 ery horizon, measurement coverage, protocols for the
006 two complementary perceptual checks, and additional
007 qualitative material. The text-conditioned and image-
008 conditioned tracks are reported here at equal depth.
009 Section numbering is independent of the main paper.

010 1. Formal Factor Definitions

011 The definitions below are those referenced from the
012 main paper’s protocol section. Notation follows it: I_t
013 is frame t of a T -frame sequence, Ω_{static} and Ω_{flow} are
014 the region masks, $\mathbf{u}_t$ is the optical flow from I_t to I_{t+1} ,
015 and $\mathcal{E}$ and $\mathcal{L}$ are the early and late windows.

016 Feature-aligned Background Drift. For matched ORB
017 pairs $\mathcal{M}_t$ between the first frame and frame t ,

$$\begin{aligned} D_{\text{feat}}(t) &= \text{median}_{(p,q) \in \mathcal{M}_t} \|p - q\|_2, \\ \text{fBD} &= \frac{100}{d} \cdot \frac{1}{|\mathcal{L}|} \sum_{t \in \mathcal{L}} D_{\text{feat}}(t) \quad (\downarrow), \end{aligned} \quad (1)$$

020 with d the frame diagonal, so fBD is a percentage of
021 image size.

Normalised Background Flow.

$$\text{NBF} = \frac{10^3}{W \Delta t} \cdot \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{|\Omega_{\text{static}}|} \sum_{\mathbf{x} \in \Omega_{\text{static}}} \|\mathbf{u}_t(\mathbf{x})\|_2 \quad (\downarrow), \quad (2)$$

022 with W the frame width and Δt the interval of the
023 sampled pairs actually used for flow estimation, so the
024 quantity is independent of both the native frame rate
025 and the sampling policy.
026

027 Global-motion compensation. With $T_t(\mathbf{x}) = s_t R_t \mathbf{x} +$
028 $\mathbf{b}_t$ the robust similarity transform fitted to static-
029 region correspondences and $\mathbf{d}_t(\mathbf{x}) = T_t(\mathbf{x}) - \mathbf{x}$ the dis-
030 placement field it induces,

$$\tilde{\mathbf{u}}_t(\mathbf{x}) = \mathbf{u}_t(\mathbf{x}) - \mathbf{d}_t(\mathbf{x}). \quad (3)$$

Motion-Compensated Foreground Flow and Flow Per-
sistence.

$$\text{MCFF}(\mathcal{W}) = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} \frac{1}{|\Omega_{\text{flow}}|} \sum_{\mathbf{x} \in \Omega_{\text{flow}}} \|\tilde{\mathbf{u}}_t(\mathbf{x})\|_2 \quad (\uparrow), \quad (4)$$

$$\text{FP} = \min \left(\frac{\text{MCFF}(\mathcal{L})}{\text{MCFF}(\mathcal{E}) + \epsilon}, c \right) \quad (\uparrow), \quad (5)$$

clipped at a fixed c so that a vanishing early window
cannot produce an unbounded ratio.

Drift leakage and attenuation. With F_{static} the mean
flow magnitude over Ω_{static} and $F_{\text{raw}}, F_{\text{comp}}$ the mean
raw and compensated magnitudes over Ω_{flow} , all in the
late window,

$$\text{DLR} = \frac{F_{\text{static}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow), \quad \text{DAR} = 1 - \frac{F_{\text{comp}}}{F_{\text{raw}} + \epsilon} \quad (\downarrow). \quad (6)$$

DLR is not a fraction and is deliberately unbounded:
DLR > 1 indicates static support moving more than
the region intended to move. DAR is stored signed
and reported clipped to $[0, 1]$.

047 2. Region Partition

Derivation. For every generated sequence we aver-
age optical-flow magnitude [8] over the leading 12%
of sampled frame pairs and threshold it by Otsu’s
method [4] into a dynamic region and its complement.
The static region is then reduced by a fixed morpholog-
ical erosion and a 4% frame border discarded; together
these form the ignored transition band, so boundary
pixels—where the partition is least certain and flow
least reliable—contribute to neither region. The same
procedure applies to both tracks.

Why the estimate is restricted to an early window.
Deriving a partition from a sequence’s entire motion
field would be circular, since accumulated background
drift would itself be read as evidence that the affected
region is dynamic. Restricting the estimate to the early
window breaks that circularity in the direction that
matters: at the 60s horizon the partition is fixed from
the first 7.1s, before long-horizon drift has accumu-
lated, and the metrics are evaluated over the remain-
der.

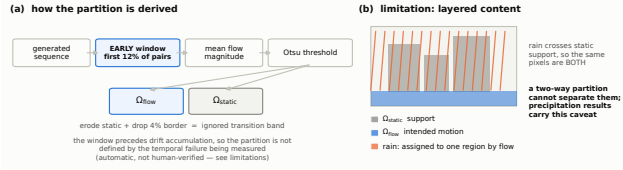


Figure 1. Region-partition protocol. (a) The partition is derived automatically from early-window flow, with an ignored transition band at the boundary. (b) A two-way partition cannot represent layered content such as rain crossing static support, which is the scope limit stated in the main paper.

Limitations. The partition is obtained automatically rather than by annotation, so no inter-annotator agreement is reported, and for the image-conditioned track it is derived per generated sequence rather than once from the shared source image. The early window is a weaker guarantee than a first-frame annotation: a system that drifts within the first 12% of its rollout can influence its own partition. We therefore treat partitions as measured quantities and report the consequence directly—every measurement is recomputed under systematic erosion and dilation of the boundary, with the response in Table 1.

3. Validation Protocol

Geometric severity is specified in a common physical unit—the mean pixel displacement a corruption induces over the frame by the end of the rollout—and each family’s parameter is derived from it using the frame geometry: a translation of d pixels, a rotation of $d/\bar{r}$ radians, or a zoom of $1 + d/\bar{r}$, where $\bar{r}$ is the mean distance from the frame centre. This is what makes the geometric families comparable. A parameter in degrees is not commensurate with one in pixels: at the evaluation resolution one degree of rotation induces roughly 2.7 px of mean displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude, and a factor can appear non-monotone simply because the corruption was too weak to clear the clip’s own baseline drift.

4. Partition and Composition Robustness

Two questions decide whether the region partition can be trusted as a measurement basis: whether a system can be advantaged by receiving an easier mask, and whether the ordering depends on the one scene category the partition cannot represent.

The static region occupies a similar share of the frame for every audited system, between 0.61 and 0.66 on average, so no system is scored over a markedly

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	drift	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25% on the factor’s target family. Rotational response is reported but not gated, and the factors divide that work explicitly: DLR is a translational-leakage diagnostic whose rotational response is inverted (−0.80), while rotational drift is carried by fBD and DAR at +0.70 and +1.00. That division is why DLR is never read alone. VBench DD is a comparison row rather than a candidate factor—it is put through the identical corruptions to show what a whole-frame score does under them—so its admission cell is left blank. Twelve reference clips per severity level.

larger or smaller support than another. Across systems, that area correlates negatively with feature-based drift. Over seven systems the correlation is not determined, but its sign is the one circularity would produce—a system that drifts within the estimation window enlarges its own dynamic region and shrinks the support its drift is measured over—and we report it as an open validity question rather than dismissing it on significance. The decisive test, recomputing every score under a region partition not derived from the same sequence, is stated as required future work in Sec. 2; we do not claim to have run it.

Removing precipitation, where rain and snow cross static support and a two-way partition must assign those pixels to one region by flow magnitude alone, leaves the normalised-background-flow ordering unchanged and moves only mid-field positions under feature-based drift.

5. Generation Configuration

Every audited system was run in its released configuration. What the run manifest records is reported below; what it does not record is stated rather than reconstructed. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were

System	static-region area
CausVid	0.663 ± 0.073
Reward-Forcing	0.661 ± 0.060
Self-Forcing	0.651 ± 0.087
Infinite-Forcing	0.650 ± 0.090
LongLive	0.644 ± 0.095
Rolling-Forcing	0.616 ± 0.107
Causal-Forcing	0.614 ± 0.086
Does mask area predict the drift score?	
Spearman(area, fBD) over 7 systems	-0.75, exact $p = 0.066$
Ordering with precipitation excluded (3 of 23 prompts)	
fBD ordering	best and worst unchanged
NBF ordering	identical

Table 2. Partition and composition robustness. The static region occupies a similar share of the frame for every audited system, so no system is scored over a much larger or smaller support than another. The area does correlate negatively with drift across systems; over seven systems that correlation is not determined, but its sign is the one circularity would produce—a system that drifts early enlarges its own dynamic region—and we flag it as an open validity question rather than dismissing it. Removing precipitation, the category a two-way partition cannot represent, leaves the normalised-background-flow ordering unchanged.

not captured at generation time, and we do not infer them after the fact. This is the weakest point in the reproducibility of the audit, and it constrains what the results support: they compare the systems as released and run here, not the space of settings each admits.

The image-conditioned roster is recorded the same way. Rows marked [†] did not reach these horizons under their released configuration and were run instead through a common long-horizon wrapper, identical for every such row: six frames per block, image-conditioned, with the released checkpoint otherwise unmodified. That wrapper is a property of this evaluation, not of any audited system, which is why its rows are read as deployment sensitivity and are never ranked against native rows. Its per-method sampler settings were not captured either, and the same limitation therefore applies.

6. Measurement Coverage

Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial. Separately, fBD is feature-based [6] and declines to report where too few repeatable keypoints survive; those abstentions are tabulated rather than folded silently into a smaller n .

7. Complete Audit, Both Tracks

The main paper reports the text-conditioned track at 60s, the tier whose sample supports method-level

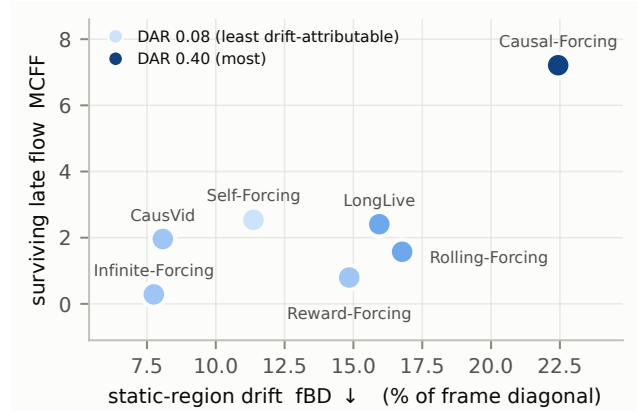


Figure 2. Operating regimes under native deployment. Each system in the static-drift–surviving-flow plane. Colour encodes the relative attenuation of measured dynamic-region flow after global-motion compensation, not a share of it attributable to drift: the quantity is signed, and values are shown clipped to $[0, 1]$. The view exposes trade-offs; it is not a leaderboard.

separation. Both tracks appear here at every horizon, with uncertainty from a percentile bootstrap over items [1]. For the image-conditioned track, systems marked matched run under a common long-horizon wrapper rather than their released configuration, so their scores are read as deployment sensitivity rather than as the published method’s performance.

System	setting	resolution	fps	horizons
CausVid	native	832×480	16	5s, 60s, 120s, 240s
Self-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Infinite-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Rolling-Forcing	native	832×480	16	5s, 60s, 120s, 240s
Reward-Forcing	native	832×480	16	5s, 60s, 120s, 240s
LongLive	native	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing	native	832×480	16	5s, 60s, 120s, 240s

Table 3. Recorded generation configuration. Every audited system was run in its released configuration and produced output at the same resolution and frame rate over the same horizons, so no comparison in this paper is confounded by output format. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured in the run manifest and are therefore not listed here; this is a reproducibility gap we state rather than fill by inference.

System	setting	resolution	fps	horizons
Causal-Forcing++ (2-step) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing++ (1-step) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing (frame-wise) [†]	matched	832×480	16	5s, 60s, 120s, 240s
Causal-Forcing++ (2-step, frame-wise)	native	832×480	16	5s, 60s, 120s, 240s
Self-Forcing [†]	matched	832×480	16	5s, 60s, 120s, 240s
CausVid [†]	matched	832×480	16	5s, 60s, 120s, 240s
Wan2.1-I2V-14B-480P	native	832×464	16	5s
Wan2.2-I2V-A14B	native	832×464	16	5s
LTX-Video 13B-0.9.8-distilled	native	832×480	24	5s
Causal-Forcing++ (1-step, frame-wise)	native	832×480	16	60s, 120s

Table 4. Recorded generation configuration. Every audited system was run in its released configuration and produced output at the same resolution and frame rate over the same horizons, so no comparison in this paper is confounded by output format. Per-method sampler settings—denoising steps, chunk length, guidance scale, cache policy—were not captured in the run manifest and are therefore not listed here; this is a reproducibility gap we state rather than fill by inference.

track	horizon	measured	fBD abstained	rate
I2V	120s	400	0	0.0%
I2V	240s	65	0	0.0%
I2V	5s	180	0	0.0%
I2V	60s	630	7	1.1%
T2V	120s	42	0	0.0%
T2V	240s	28	0	0.0%
T2V	5s	48	0	0.0%
T2V	60s	185	0	0.0%

Table 5. Measurement coverage and fBD abstention. fBD is feature-based and declines to report when too few repeatable keypoints survive in the static region. Every abstention in this benchmark occurs on a single desert dust-storm scene, whose flat, dust-obscured ground offers no stable structure to track, and it occurs there for every system alike—so it reflects the scene, not a method. The remaining factors are reported for those clips as usual; only fBD is withheld.

8. Aggregation Sensitivity

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The evaluation set is dominated by one flow medium, so the audit must be shown not to be a report on that medium alone. The main paper aggregates by averaging over prompts. The alternative, weighting each scene category equally, is not preferable here: at the principal horizon three of the five populated categories contain two prompts each, so a macro-average would give a two-prompt cell the same influence as a fourteen-prompt one and inflate variance in exchange for nominal balance. We therefore report both and state what changes.

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The best- and worst-ranked systems are the same under either aggregation, so the contrast the paper draws does not depend on the choice. Intermediate ranks do move, which is why no ordinal claim is made about the middle of the field.

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Method	status	setting	5s	60s	120s	240s
CausVid	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Self-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Infinite-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Rolling-Forcing	public	native	V6 T6 B	V24 T24 B	V6 T6 B	V4 T4 B
Reward-Forcing	public	native	V12 T12 B	V23 T23 B	V6 T6 B	V4 T4 B
LongLive	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B
Causal-Forcing	public	native	V6 T6 B	V23 T23 B	V6 T6 B	V4 T4 B

Table 6. Asset and score coverage for the audited public systems. Tn/N marks entries where a metrics file exists but only n of N videos hold usable measurements.

Method	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 6$)							
CausVid	2.08	6.56	1.44	2.91	1.24	0.26	0.07
Self-Forcing	0.68	10.8	7.70	4.35	0.76	0.49	0.03
Infinite-Forcing	0.24	2.32	1.23	0.89	1.08	0.21	0.09
Rolling-Forcing	3.51	12.1	3.10	3.69	1.32	0.48	0.17
Reward-Forcing	1.10	5.21	0.63	0.58	1.37	0.44	0.28
LongLive	1.91	7.31	1.25	2.27	1.16	0.59	0.32
Causal-Forcing	9.55	31.0	5.32	3.01	1.13	0.62	0.37
60s horizon ($n = 23$)							
CausVid	8.08	9.27	2.87	1.96	0.67	0.45	0.14
Self-Forcing	11.4	12.7	10.4	2.54	0.44	0.50	0.14
Infinite-Forcing	7.75	3.65	1.17	0.29	0.63	0.55	0.14
Rolling-Forcing	16.8	11.2	3.18	1.57	0.75	0.54	0.18
Reward-Forcing	14.8	8.51	3.81	0.80	0.49	0.60	0.17
LongLive	15.9	13.9	6.46	2.41	0.75	0.59	0.17
Causal-Forcing	22.4	87.6	9.84	7.21	0.83	0.94	0.43
120s horizon ($n = 6$)							
CausVid	6.15	3.42	0.36	0.52	1.06	0.41	0.13
Self-Forcing	21.2	6.80	4.85	0.82	0.85	0.75	0.15
Infinite-Forcing	16.1	2.60	0.50	0.32	0.64	0.57	0.21
Rolling-Forcing	23.8	10.6	3.33	0.77	0.38	0.82	0.22
Reward-Forcing	20.0	11.6	2.05	0.69	0.39	0.70	0.12
LongLive	22.5	9.66	5.10	1.19	0.81	0.84	0.23
Causal-Forcing	21.8	130	10.5	10.7	0.90	0.81	0.50
240s horizon ($n = 4$)							
CausVid	7.23	13.4	0.42	0.87	0.99	0.85	0.08
Self-Forcing	20.0	17.4	3.09	0.50	0.15	2.26	0.20
Infinite-Forcing	20.0	4.15	0.33	0.23	1.02	1.08	0.15
Rolling-Forcing	23.7	15.4	2.15	0.97	0.46	0.78	0.18
Reward-Forcing	26.4	11.4	1.09	3.55	1.53	0.50	0.07
LongLive	23.1	11.8	1.52	1.32	1.20	1.38	0.31
Causal-Forcing	18.9	102	4.72	6.90	1.21	0.73	0.58

Table 7. Complete T2V audit, all four horizons. The main paper reports the 60s and 120s panels; the 5s tier is an initialisation check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. All systems are evaluated in the native setting. n is stated per horizon and is common to every row of that panel.

9. Whole-Frame and SNF-Bench Rank Correlation 181
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10. Deployment Sensitivity 183

11. Signed Drift Attenuation 184

5 Drift attenuation is stored signed and reported clipped to $[0, 1]$. Negative values are informative rather than 185
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Method	setting	fBD↓	NBF↓	MCFF-E	MCFF-L↑	FP↑	DLR↓	DAR↓
5s horizon ($n = 10$)								
Causal-Forcing++ (2-step)	matched	3.69	10.0	1.06	0.61	0.78	0.67	0.32
Causal-Forcing++ (1-step)	matched	5.28	11.4	1.05	0.47	0.64	0.66	0.29
Causal-Forcing (frame-wise)	matched	16.1	45.3	8.48	2.79	0.43	0.68	0.29
Causal-Forcing++ (2-step, frame-wise)	native	1.22	10.6	2.23	0.78	0.47	0.68	0.20
Self-Forcing	matched	9.66	5.74	0.69	0.26	0.70	0.74	0.21
CausVid	matched	15.2	10.8	0.76	0.87	0.98	1.04	0.10
Wan2.1-I2V-14B-480P	native	5.01	21.3	6.67	5.58	1.09	0.52	0.18
Wan2.2-I2V-A14B	native	6.31	25.3	4.33	9.09	0.95	0.50	0.27
LTX-Video 13B-0.9.8-distilled	native	6.65	17.8	3.59	3.48	0.92	0.39	0.24
60s horizon ($n = 30$)								
Causal-Forcing++ (2-step)	matched	9.18	5.97	0.85	0.60	0.86	0.67	0.25
Causal-Forcing++ (1-step)	matched	5.34	7.05	0.86	0.57	0.77	0.64	0.28
Causal-Forcing (frame-wise)	matched	21.1	36.4	2.80	2.69	0.89	0.80	0.35
Causal-Forcing++ (2-step, frame-wise)	native	9.94	6.91	1.30	0.78	1.02	0.75	0.17
Self-Forcing	matched	17.3	5.52	1.50	0.69	0.89	0.67	0.18
CausVid	matched	19.9	3.31	0.58	0.41	0.81	0.57	0.14
Causal-Forcing++ (1-step, frame-wise)	native	10.4	5.25	0.53	0.52	1.01	0.83	0.22
120s horizon ($n = 20$)								
Causal-Forcing++ (2-step)	matched	9.65	8.81	0.89	0.82	0.93	0.58	0.28
Causal-Forcing++ (1-step)	matched	7.26	9.39	0.84	0.66	0.86	0.62	0.31
Causal-Forcing (frame-wise)	matched	19.9	52.7	4.82	5.06	1.08	0.67	0.24
Causal-Forcing++ (2-step, frame-wise)	native	11.4	17.3	1.89	1.54	1.01	0.75	0.09
Self-Forcing	matched	23.0	10.3	2.41	0.76	0.45	0.66	0.26
CausVid	matched	19.7	6.04	1.15	0.83	0.67	0.48	0.14
Causal-Forcing++ (1-step, frame-wise)	native	13.6	7.39	1.01	0.35	0.64	0.77	0.33
240s horizon ($n = 5$)								
Causal-Forcing++ (2-step)	matched	10.4	4.83	0.53	0.59	1.05	0.55	0.15
Causal-Forcing++ (1-step)	matched	6.42	6.66	0.56	0.51	0.83	0.67	0.30
Causal-Forcing (frame-wise)	matched	19.8	31.5	3.08	2.28	0.74	0.69	0.45
Causal-Forcing++ (2-step, frame-wise)	native	6.32	3.95	1.23	0.68	0.67	0.83	0.02
Self-Forcing	matched	22.3	5.74	1.48	1.59	0.81	0.71	0.08
CausVid	matched	18.4	2.08	0.81	0.79	0.86	0.44	0.11

Table 8. Complete I2V audit, all four horizons. The main paper reports the 60s and 120s panels; the 5s tier is an initialisation check and 240s a diagnostic extreme. Every row is computed under the same frozen specification, so panels are comparable within a horizon; prompt sets differ across horizons, so columns are not comparable between panels. n is stated per horizon and is common to every row of that panel.

187 anomalous: where local flow opposes the estimated
 188 global field, compensation increases the measured mag-
 189 nitude. We report their incidence rather than suppress-
 190 ing it.

191 12. Evaluation-Set Composition

192 13. Exploratory Perceptual Pilot

193 This section supports no perceptual-validity claim. It
 194 reports an exploratory pilot that did not establish
 195 agreement between human judgments and the factor
 196 orderings, and it is included because the reason it failed
 197 is informative and because the protocol is what a pow-

ered study should start from. Nothing here ranks sys-
 tems, and no result from it enters any factor, table or
 figure elsewhere in this work.

The question the pilot asked is whether independent
 judges order clip pairs the way the factors do. That
 is metric validation rather than preference, so a small
 sample would have been survivable had the pairs been
 constructed correctly; as Sec. 13 reports below, they
 were not.

Ethics. Raters were volunteers drawn from the au-
 thors' institution and immediate colleagues, and four of
 the eight sessions were the authors' own pilot runs, re-
 ported separately and excluded from the primary anal-

System	prompt mean	rank	macro mean	rank
NBF				
Infinite-Forcing	3.65	1	4.12	1
Reward-Forcing	8.51	2	10.11	4 [†]
CausVid	9.27	3	9.93	3
Rolling-Forcing	11.23	4	11.32	5 [†]
Self-Forcing	12.72	5	8.98	2 [†]
LongLive	13.86	6	14.69	6
Causal-Forcing	87.62	7	61.70	7
fBD				
Infinite-Forcing	7.75	1	6.10	1
CausVid	8.08	2	9.51	3 [†]
Self-Forcing	11.37	3	9.27	2 [†]
Reward-Forcing	14.85	4	15.61	5 [†]
LongLive	15.94	5	14.93	4 [†]
Rolling-Forcing	16.77	6	18.85	6
Causal-Forcing	22.44	7	23.49	7

Table 9. Sensitivity of the audit to category aggregation. Prompt means, as reported in the main paper, against category macro-averages, for the two static-fidelity factors at the principal horizon. [†] marks a system whose rank changes. The best- and worst-ranked systems are identical under both, so the paper’s headline contrast does not depend on the choice; intermediate positions do change, which is why no ordinal claim is made about the middle of the field.

generic metric	fBD	NBF	MCFF	DLR
VB-DD	+0.75	+0.93	+0.89	+0.36
VB-bg	-0.64	-0.71	-0.75	-0.50
VB-smooth / VB-flick	-0.96	-0.82	-0.61	-0.64

Table 10. Rank disagreement at 60s. Spearman correlation over $n = 7$ method means—descriptive, not inferential. Positive correlation between apparent motion and background flow is the effect the benchmark isolates. Smoothness and flickering rank these systems identically and share a row.

211 ysis. Participation was voluntary and uncompensated,
212 no personal data beyond a self-chosen identifier was
213 collected, and no clip depicts an identifiable person.
214 The study involved no intervention and no sensitive
215 material. Because a powered study will recruit outside
216 this group, we will obtain the institutional review or
217 documented exemption that such recruitment requires
218 before running it.

219 Questions. One forced choice per axis, with an ex-
220 plicit third option: static stability—in which clip does
221 the background stay more fixed in place—and motion
222 persistence—in which clip does the intended motion
223 keep going rather than fade or freeze. Judges are never
224 asked which output is better, or which looks more nat-
225 ural. “Better” conflates the two failures the benchmark
226 separates, and naturalness is outside the stated scope,
227 so either question would produce a result the paper

could not own. 228

Pairs. A pair is two systems’ outputs on the same 229
prompt at the same horizon, never across prompts. 230
Pairs are stratified by the relative gap in the axis’s own 231
factor: clear-gap pairs, where the factor separates the 232
two outputs, carry the agreement claim, while near-tie 233
pairs test the opposite property—that where the fac- 234
tors do not separate, judges also cannot tell the clips 235
apart. Presentation side is randomised per trial and re- 236
solved back to a system identity only after the response 237
is recorded; system names are never shown. 238

Statistics. The unit of analysis is the pair, majority- 239
voted across raters, and intervals are percentile boot- 240
strap resampling pairs. Responses are clustered inside 241
rater and inside pair, so an interval computed over re- 242
sponses would claim precision the design does not have. 243
We report inter-rater agreement so the reader can see 244
whether the raters are signal or noise, and we run no 245
significance test the sample cannot power: the claim is 246
descriptive agreement, and saying so plainly is prefer- 247
able to a p -value that would not survive scrutiny. 248

Outcome of the human arm: the instrument works, 249
agreement is not yet estimable. Eight raters each 250
completed all 24 trials, four of them independent of the 251
authors; the remaining four were pilot sessions and are 252
reported separately rather than pooled. Engagement 253
was genuine—median deliberation ran from 23 to 233 254

Method	Generic evidence	SNF-Bench evidence	Interpretation
CausVid	VB-DD rank 6/7	NBF rank 3/7, MCFF-L rank 4/7	stable sequence with decaying intended motion
Causal-Forcing	VB-DD rank 1/7	NBF rank 7/7, MCFF-L rank 1/7	substantial apparent motion, strongly contaminated by background displacement
Infinite-Forcing	VB-DD rank 7/7	NBF rank 1/7, MCFF-L rank 7/7	stable sequence with decaying intended motion
LongLive	VB-DD rank 3/7	NBF rank 6/7, MCFF-L rank 3/7	substantial apparent motion, strongly contaminated by background displacement
Reward-Forcing	VB-DD rank 5/7	NBF rank 2/7, MCFF-L rank 6/7	stable sequence with decaying intended motion
Self-Forcing	VB-DD rank 2/7	NBF rank 5/7, MCFF-L rank 2/7	substantial apparent motion, strongly contaminated by background displacement

Table 11. Interpretation changes between generic whole-frame metrics and SNF-Bench at 60s. Neither evaluation is labelled correct; the table identifies information hidden by whole-frame aggregation. Cases are selected by a fixed rank-gap rule applied to all public methods, so none can be selectively omitted.

Checkpoint	horizon	Δ fBD	Δ NBF	Δ MCFF-L	Δ DLR
Causal-Forcing++ (2-step)	5s	+2.47	-0.61	-0.17	-0.01
Causal-Forcing++ (2-step)	60s	-0.76	-0.95	-0.17	-0.08
Causal-Forcing++ (2-step)	120s	-1.71	-8.46	-0.73	-0.17
Causal-Forcing++ (2-step)	240s	+4.06	+0.88	-0.09	-0.28
Causal-Forcing++ (1-step)	60s	-5.05	+1.80	+0.05	-0.20
Causal-Forcing++ (1-step)	120s	-6.34	+2.00	+0.31	-0.14

Table 12. Deployment sensitivity. Change from each checkpoint’s native configuration to the common long-horizon wrapper, for the two checkpoints evaluated both ways. Absolute matched scores are never read as the published method’s performance.

Method	track	dur	n	negative	rate	most negative
CausVid	i2v	120s	20	1	5.0%	-0.055
CausVid	i2v	5s	10	5	50.0%	-1.076
CausVid	i2v	60s	30	2	6.7%	-0.084
Causal-Forcing++ (1-step)	i2v	120s	20	1	5.0%	-0.027
Causal-Forcing (frame-wise)	i2v	120s	20	3	15.0%	-0.171
Causal-Forcing++ (2-step)	i2v	5s	10	1	10.0%	-0.164
Causal-Forcing++ (1-step, frame-wise)	i2v	60s	30	3	10.0%	-1.420
Causal-Forcing++ (2-step, frame-wise)	i2v	120s	20	5	25.0%	-0.241
Causal-Forcing++ (2-step, frame-wise)	i2v	240s	5	1	20.0%	-0.651
Causal-Forcing++ (2-step, frame-wise)	i2v	5s	10	1	10.0%	-0.288
Causal-Forcing++ (2-step, frame-wise)	i2v	60s	30	1	3.3%	-0.001
LTX-Video 13B-0.9.8-distilled	i2v	5s	10	2	20.0%	-0.018
Self-Forcing	i2v	120s	20	2	10.0%	-0.046
Self-Forcing	i2v	240s	5	1	20.0%	-0.136
Self-Forcing	i2v	60s	30	4	13.3%	-0.578
Wan2.1-I2V-14B-480P	i2v	5s	10	1	10.0%	-0.255
Wan2.2-I2V-A14B	i2v	5s	10	1	10.0%	-0.335
CausVid	t2v	240s	4	1	25.0%	-0.384
CausVid	t2v	5s	6	2	33.3%	-0.093
Causal-Forcing	t2v	5s	6	2	33.3%	-0.273
Causal-Forcing	t2v	60s	23	3	13.0%	-0.640
Infinite-Forcing	t2v	240s	4	1	25.0%	-0.158
Infinite-Forcing	t2v	60s	23	4	17.4%	-0.359
LongLive	t2v	120s	6	1	16.7%	-1.076
LongLive	t2v	5s	6	1	16.7%	-0.139
LongLive	t2v	60s	23	2	8.7%	-0.051
Reward-Forcing	t2v	120s	6	1	16.7%	-0.116
Reward-Forcing	t2v	5s	12	2	16.7%	-0.145
Reward-Forcing	t2v	60s	23	3	13.0%	-0.317
Rolling-Forcing	t2v	120s	6	3	50.0%	-0.353
Rolling-Forcing	t2v	5s	6	1	16.7%	-0.068
Rolling-Forcing	t2v	60s	24	2	8.3%	-0.038
Self-Forcing	t2v	120s	6	2	33.3%	-0.130
Self-Forcing	t2v	240s	4	1	25.0%	-0.153
Self-Forcing	t2v	5s	6	1	16.7%	-0.046
Self-Forcing	t2v	60s	23	4	17.4%	-1.117

Table 13. Incidence of negative DAR, i.e. clips where global-motion compensation increased measured dynamic-region flow. 21 further entries had no negative values and are omitted.

seconds per trial—and the “can’t tell” option was used deliberately, on 30% of responses, rather than avoided.

The agreement estimate, however, does not survive its own error bars. Majority voting across raters leaves

three decided pairs on the static axis and five on the persistence axis, and the corresponding intervals span most of the unit range. Inter-rater agreement is low ($\alpha = 0.18$ and 0.24 on the two axes), and the direc-

track/duration	n	river_stream	ocean_waves	precipitation	fire_smoke	lava_volcanic	windborne	cats
i2v/5s	10	2	.	3	2	1	2	5
i2v/60s	30	9	4	9	1	3	4	6
i2v/120s	20	6	3	3	3	1	4	6
i2v/240s	5	1	2	1	.	1	.	4
t2v/5s	6	3	1	1	1	.	.	4
t2v/60s	23	14	2	3	2	.	2	5
t2v/120s	6	4	.	1	.	.	1	3
t2v/240s	4	2	.	1	.	.	1	3

Table 14. Scene-category balance of the evaluation set, by track and horizon. A dot marks a category absent from that cell.

Axis	Judge	agrees with factor	called indistinguishable	pairs	inter-rater α
Static stability (fBD)	human	– (3 decided)	–	8	0.18
	vision–language	– (1 decided)	1/1	7	–
Motion persistence (MCFF-L)	human	80% [40, 100]	–	8	0.24
	vision–language	– (2 decided)	–	3	–

Table 15. Preliminary perceptual validation of the factor axes. Each judge answers one question per axis and never which output is better. agree is agreement with the factor’s ordering on pairs the factor separates clearly, over pairs the judge decided; can’t tell is the share of near-tie pairs called indistinguishable, which is the correct answer there. The unit is the pair, majority-voted across raters, and the interval is a percentile bootstrap resampling pairs rather than responses, which are clustered within rater and pair. A dash marks a cell with too few decided pairs to carry a rate; the count is given instead. This is a preliminary check on a measurement instrument, not a powered study, and no system is ranked by it.

263 tion of the estimate reverses depending on which rater
 264 subset is used. We therefore report the counts and the
 265 interval rather than a headline percentage, and we do
 266 not claim that perceptual agreement has been estab-
 267 lished.

268 The cause is a property of the pair set rather than of
 269 the raters, and it is measurable. The pairs were strat-
 270 ified by the gap in each axis’s own factor but were not
 271 isolated against the other axis. On the static ques-
 272 tion the median relative gap in the target factor is
 273 0.29 while the median gap in the persistence factor is
 274 0.58: the clips shown differ about twice as much on the
 275 axis the rater was told to ignore as on the one being
 276 asked about, and on half the pairs both axes differ sub-
 277 stantially. A rater seeing one clip drift and the other
 278 stall cannot report which axis drove the answer, which
 279 is exactly the condition that produces low inter-rater
 280 agreement and near-chance ordering. A powered study
 281 therefore requires axis-isolated pairs—large gap in the
 282 target factor, small gap in the other—and that con-
 283 straint, not rater count, is the first thing to fix.

284 Outcome of the vision–language arm: negative, with a
 285 located cause. The same pairs and the same wording
 286 were put to an open-weight judge [5] at temperature
 287 zero, each pair asked in both presentation orders. Two
 288 controls precede any reading of its answers. Shown one
 289 clip against itself, where the only correct answer is that
 290 they are the same, it answered correctly on six of six.
 291 Shown a frozen single frame against a real sequence—

292 the largest difference available on the motion axis—its
 293 verdict followed the swapped clip on only two of six.
 294 Presentation bias is absent: over 48 queries the two
 295 slots were chosen 8 and 12 times. The judge is therefore
 296 well behaved and still cannot do the task: it answered
 297 “indistinguishable” on most pairs the factors separate
 298 clearly, leaving one and two decided pairs on the two
 299 axes, which is far too few to support an agreement rate.
 300 We report none.

301 The controls locate the cause in the judge rather
 302 than in the factors or the prompt. A model that can-
 303 not separate a still image from moving water is not
 304 positioned to separate two rollouts that differ subtly.
 305 This says that an open-weight judge at this scale can-
 306 not presently substitute for human raters on these axes.
 307 It does not say that stronger judges would fail, and we
 308 make no claim about them: contemporary generation
 309 benchmarks that report successful automated judging
 310 use frontier proprietary models [2, 7] or scorers fine-
 311 tuned for the task [3], and we have no evidence either
 312 way about those here.

313 Status. Both arms are reported as preliminary and
 314 neither supports an agreement claim. What the pilot
 315 does establish is that the instrument runs end to end,
 316 that raters engage with the axis questions rather than
 317 defaulting to a side, and that the binding constraint
 318 on a powered study is pair construction rather than
 319 rater count: axis-isolated pairs, a near-tie stratum, and
 320 raters recruited outside the authors. Protocol, pair

321 manifest and judge prompt are released verbatim so
322 the check can be repeated and enlarged. To restate
323 the scope without qualification: what is released here
324 is a validated protocol and a diagnosis of why this run
325 of it failed. No conclusion about perceptual agreement,
326 in either direction, should be drawn from the counts in
327 this section. They are reported so the pilot is auditable,
328 not because they support an inference.

329 14. Additional Qualitative Material

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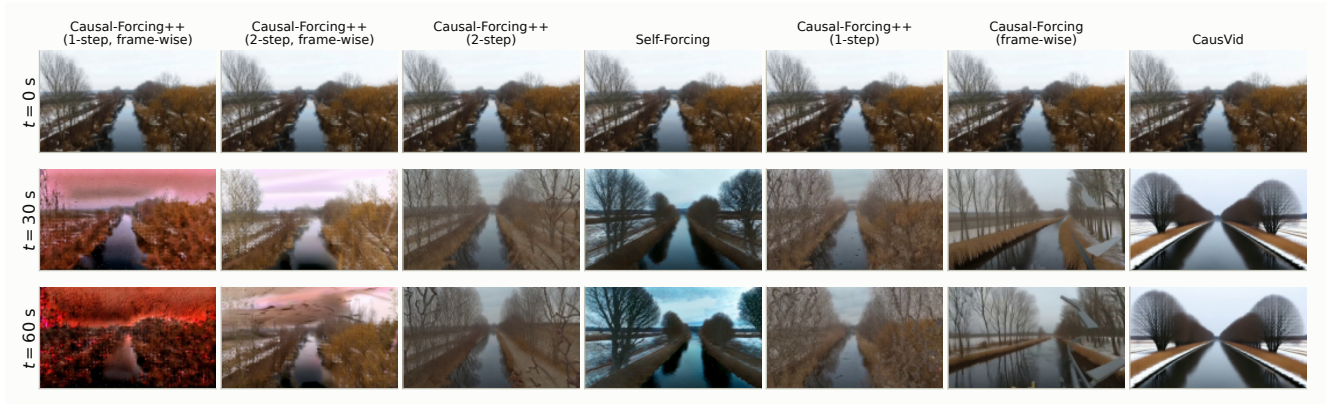


Figure 3. Image-conditioned systems on one shared source image. Every system begins from the identical frame, so divergence over the rollout is attributable to the system rather than to a differently imagined scene—which is not true of the text-conditioned track, where each system authors its own layout. Columns are ordered by increasing static-region drift.

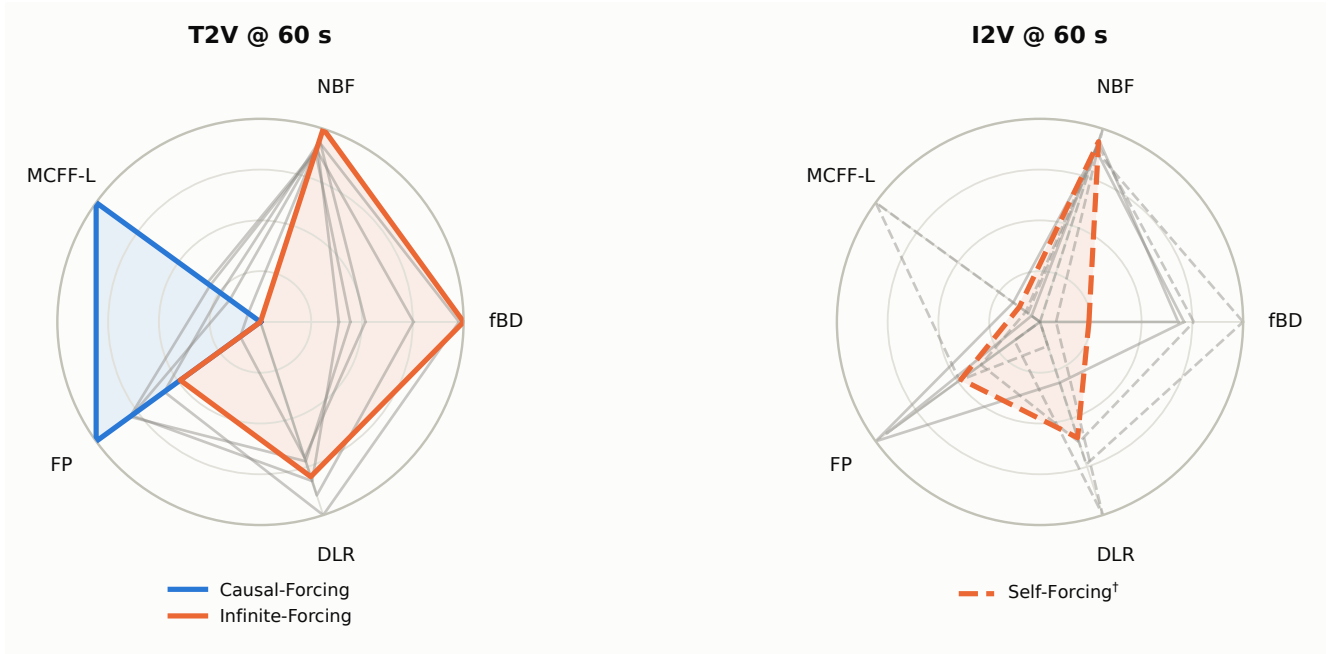


Figure 4. Factor profile per track. Each factor is rescaled across the audited systems and oriented so that outward is always better; plotting raw values would place “most drift” and “most surviving motion” both outward and the shape would carry no meaning. No system encloses the others on either track, which is the conclusion the operating-regime view reaches, shown as a profile. On the image-conditioned panel, dashed outlines marked [†] are rows run under the common long-horizon wrapper rather than a released configuration: they are shown for shape, and are not peers of the solid native outlines. The rescaling is relative to the audited roster, so a profile is descriptive of this set and would change if the set changed.

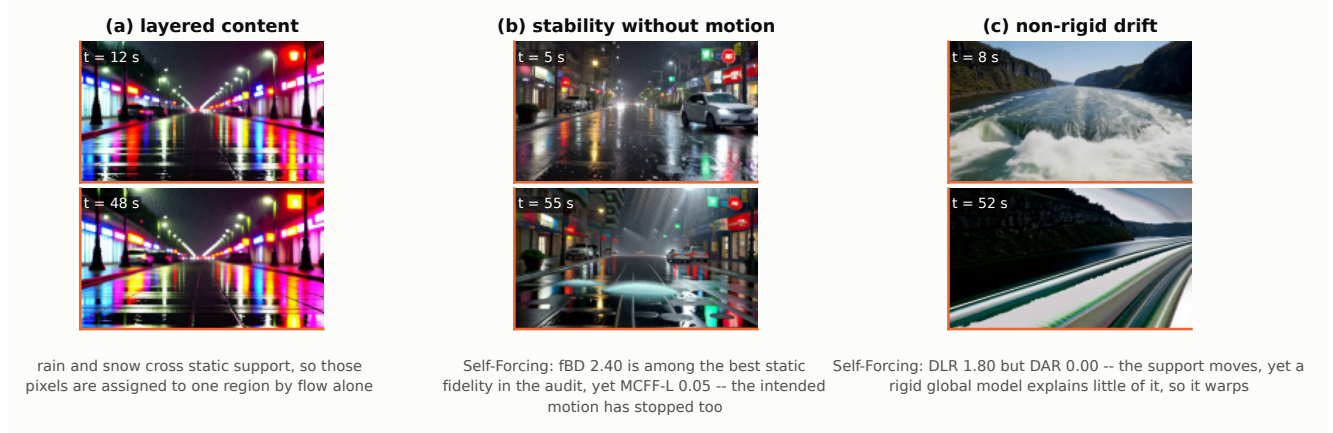
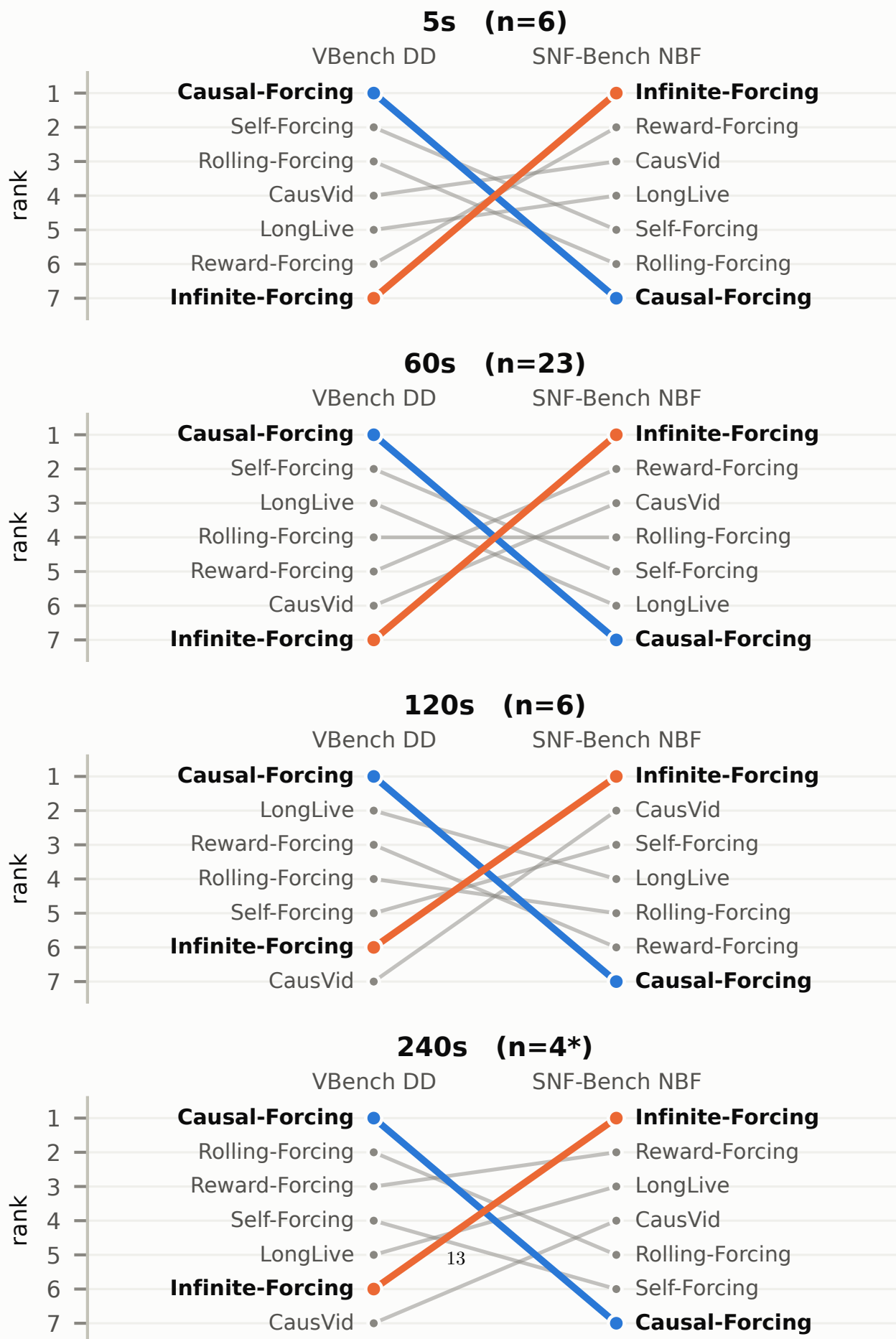


Figure 5. Stated scope boundaries. (a) A two-way partition cannot separate precipitation from the support it crosses. (b) A system can attain near-best static fidelity precisely because its intended motion has also stopped. (c) High leakage with low attenuation identifies support that moves without translating, which a rigid global model does not explain.



Figure 6. Every audited text-conditioned system on one prompt. Columns are ordered by increasing static-region drift; the two highlighted systems are the pair a whole-frame motion score cannot separate.



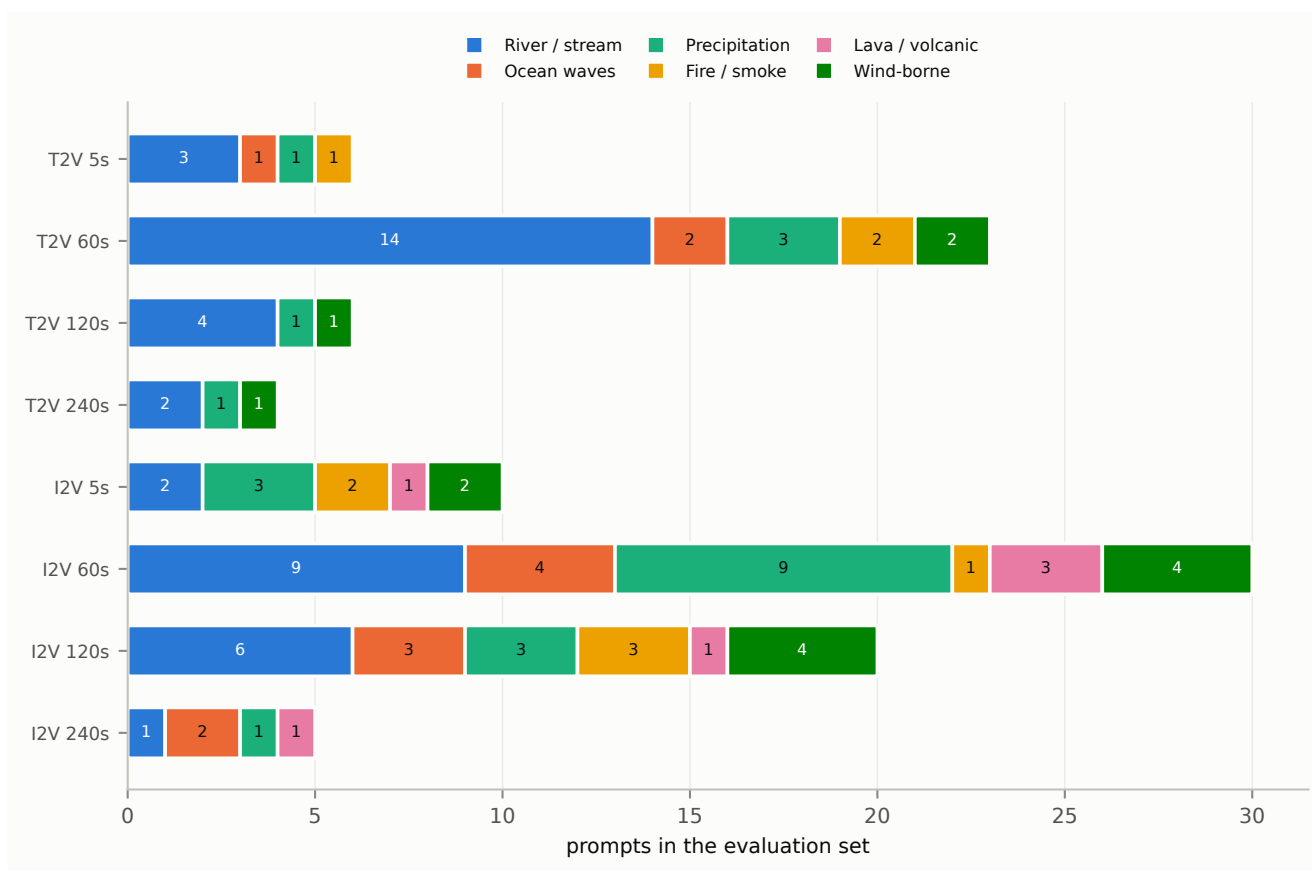


Figure 8. Composition of the evaluation set by scene category, in prompts per category at each track and horizon.